Six-Step Operation of PMSM With Instantaneous Current Control

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***Abstract*—Six-step operation has many advantages in permanent-magnet synchronous machine (PMSM) drives such as maximum power utilization and widened flux-weakening region. However, due to the maximum utilization of inverter output, saturation of current regulator makes it difficult to maintain instantaneous current control capability. Accordingly, in most of conventional research studies, the six-step operation has been implemented by voltage angle control without current regulation, whose dynamic performance is quite unsatisfactory. This paper proposes a control scheme for the six-step operation of PMSM with enhanced dynamic performance of current control. By collaborative operation of dynamic overmodulation, flux weakening, and a technique for enhanced dynamic performance, the six-step operation is realized without losing instantaneous current control capability. Simulations and experiments are carried out to verify the effectiveness of the proposed control scheme. The experimental results show 17% extension of the constant torque region and 27% enhancement of torque capability at three times of the base speed.**

***Index Terms*—Current control, flux weakening, motor drives, overmodulation, permanent-magnet synchronous machines (PMSMs), six-step operation.**

NOMENCLATURE

Superscript ^{s}	Stationary reference frame.
Superscript ^{r}	Synchronously rotating rotor reference frame.
Superscript *	Reference value.
P	Number of poles.
ω_r	Electrical rotor speed.
V_s^*	Magnitude of voltage reference.
V_{s1}	Fundamental component of inverter output voltage.
v_{ds}^r, v_{qs}^r	$d-q$ components of inverter output in rotor reference frame.
i_{ds}^r, i_{qs}^r	$d-q$ components of phase current in rotor reference frame.
R_s	Stator resistance.
L_{ds}, L_{qs}	$d-q$ components of stator self-inductance.
ΔL_s	Difference between $d-q$ inductances $L_{ds} - L_{qs}$.
λ_f	Back electromotive force (EMF) constant.
T_e	Output torque of permanent-magnet synchronous machine (PMSM).
$V_{\rm dc}$	DC-link voltage.

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I. INTRODUCTION

BACK EMF voltage of a PMSM is proportional to the speed of the rotor. To keep the current regulation capability, the output voltage of the inverter should also increase proportionally to the speed. However, the output voltage of the inverter is limited by DC-link voltage, which makes physical limitation of operating speed of the machine. In order to overcome the voltage constraint in PMSM drives, flux-weakening control, which applies negative d-axis current to reduce d-axis flux linkage, is normally used. In the early research studies of the flux-weakening control of PMSM [1], [2], only linear modulation region is considered for the steady-state operation. Although dynamic performance would be fine with these methods, they do not fully utilize the inverter capability.

Six-step operation of the inverter has been reviewed by many researchers for its strong merit that the fundamental component of the inverter output voltage can be maximized [3]–[13]. That is, by the six-step operation, feasible inverter output voltage can be enhanced by 10.27%. Accordingly, torque capability and operation region of PMSM can be extended without any modification of hardware. Although the six-step operation causes subproblems such as torque ripple and acoustic noise, it has many advantages in traction drives for electric vehicle and hybrid electric vehicle applications where the average torque at the higher speed is premium under given battery voltage.

Although, as aforementioned, the six-step operation has many advantages, maintaining high dynamic performance of current control during the six-step operation is a challenging task. Conventional vector control only works under the premise that both the amplitude and the phase angle of the inverter output are adjustable. However, during the six-step operation, there is only one degree of freedom for the inverter output: the phase angle of the voltage. Conventional vector control strategy is incompatible with the six-step operation.

In the conventional studies about the six-step operation [3]–[13], dynamic control of current had not been addressed. References [3]–[7] mainly focused on voltage synthesis itself of the inverter. However, only steady-state operation had been considered in these research studies, neglecting transient-state behavior of current. References [8]–[13] proposed voltage

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angle control methods. In these research studies, closed-loop current regulator was used only under the base speed. At higher speed, control algorithm was switched to the voltage angle control deactivating the current regulator. This kind of control algorithm transition may cause shock or hunting to the drive system. Moreover, the voltage angle control itself has limited dynamic performance due to its inherent nature of open-loop control.

In order to achieve the instantaneous current control capability, closed-loop current regulator is needed. Some authors proposed flux-weakening controllers incorporated with current regulators, operating with nonlinear modulation of voltage [14]–[18]. In [14] and [15], flux-weakening algorithms operating with saturated current regulator was proposed. In the algorithm, current control capability was achieved by proper modification of the current references by low-pass-filtered voltage error signal. References [16] and [17] also proposed flux-weakening controllers based on low-pass filter (LPF). Reference [18] proposed a flux-weakening controller using active switching time signal feedback. Although both high degree of voltage utilization and current control capability had been achieved by [14]–[18], they could not cover the six-step operation where the inverter capability is maximally utilized.

This paper proposes a control scheme for the six-step operation of PMSM with instantaneous current control. In the proposed scheme, a dynamic overmodulation method, a flux-weakening controller, and a technique for enhancing dynamic performance are working together with the closed-loop current regulator. In the proposed scheme, the six-step operation is realized by letting the voltage reference rotate outside the voltage hexagon and producing the six-step voltage by dynamic overmodulation. Contrary to the voltage angle control, there is no switching between different algorithms. That is, all controllers are working together in whole operating range, including the six-step operation region. In addition, instantaneous current control capability is still maintained during the six-step operation. Simulations and experiments are carried out to verify the effectiveness of the proposed method.

II. Voltage Angle Control for PMSM Drives

This section addresses the basic principle of conventional voltage angle control for PMSM drives and analysis of its dynamic performance. A mathematical model of PMSM is shown as follows:

$$\mathbf{v_s^r} = \begin{bmatrix} v_{ds}^r \\ v_{qs}^r \end{bmatrix} = \begin{bmatrix} R_s + sL_{ds} & -\omega_r L_{qs} \\ \omega_r L_{ds} & R_s + sL_{qs} \end{bmatrix} \begin{bmatrix} i_{ds}^r \\ i_{qs}^r \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_r \lambda_f \end{bmatrix} \quad (1)$$

$$T_e = \frac{3}{2}\frac{P}{2}\left\{\lambda_f i_{qs}^r + \Delta L_s \cdot i_{ds}^r i_{qs}^r\right\}. \quad (2)$$

Neglecting inductive and resistive voltage drop terms, steady-state current can be approximated as

$$i_{ds}^r = \frac{v_{qs}^r}{L_{ds}\omega_r} - \frac{\lambda_f}{L_{ds}}$$
$$i_{qs}^r = -\frac{v_{ds}^r}{L_{qs}\omega_r}. \quad (3)$$

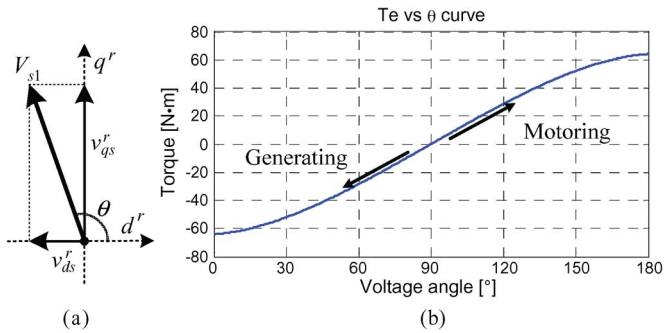


Fig. 1. (a) Definition of V_m and θ. (b) T_e versus θ curve.

Substituting (3) into (2), output torque can be deduced as

$$T_e = -\frac{3P}{4L_{ds}L_{qs}\omega_r^2}\left\{L_{qs}\omega_r\lambda_f v_{ds}^r + \Delta L_s v_{qs}^r v_{ds}^r\right\}. \quad (4)$$

Another way to express torque in terms of voltage magnitude and angle is

$$T_e = -\frac{3P}{8L_{ds}L_{qs}\omega_r^2}\left\{2L_{qs}\omega_r\lambda_f \cdot V_{s1}\cos\theta + \Delta L_s \cdot V_{s1}^2 \sin 2\theta\right\}. \quad (5)$$

V_{s1} and θ are denoted in Fig. 1(a) under the assumption of the steady-state operation. Note that V_{s1} is the fundamental component of the inverter output. To PMSM, V_{s1} is the fundamental component of the input terminal voltage. Under the six-step operation, V_{s1} is saturated at $2/\pi V_{\rm dc}$ [7]. Equation (5) claims that the output torque can be controlled by only adjusting θ when V_{s1} is saturated. The torque versus voltage angle curve of the experimental motor specified in Table II at 1500 r/min is shown in Fig. 1(b).

Conventional voltage angle control algorithms proposed in [8]–[13] adjust θ for the torque control. The voltage angle θ can be directly calculated from (5) with given torque reference and rotating speed, or it can be adjusted by a proportional–integral (PI) regulator. In any case, the closed-loop current regulator was deactivated, which means that the actual dynamic relation between the voltages and the currents is neglected. Full expression of the currents considering all transient terms can be deduced as

$$\begin{bmatrix} i_{ds}^r \\ i_{qs}^r \end{bmatrix} = \frac{1}{D(s)}\begin{bmatrix} R_s + sL_{qs} & \omega_r L_{qs} \\ -\omega_r L_{ds} & R_s + sL_{ds} \end{bmatrix}\begin{bmatrix} v_{ds}^r \\ v_{qs}^r \end{bmatrix} - \frac{1}{D(s)}\begin{bmatrix} \omega_r L_{qs} \\ R_s + sL_{ds} \end{bmatrix}\omega_r\lambda_f \quad (6)$$

where

$$D(s) = s^2 L_{ds}L_{qs} + sR_s(L_{ds} + L_{qs}) + R_s^2 + \omega_r^2 L_{ds}L_{qs}. \quad (7)$$

As shown in (6), $d-q$ currents are expressed by second-order transfer functions from $d-q$ voltages. If $d-q$ voltages are simply changed in a step manner, the current response would

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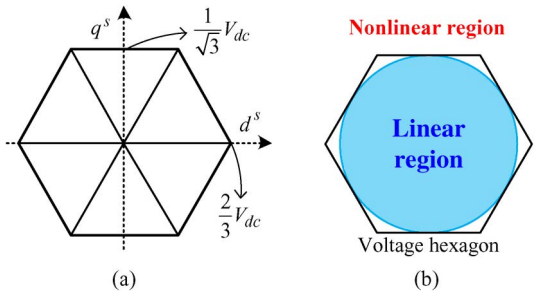


Fig. 2. (a) Voltage hexagon and its (b) division for voltage synthesis.

be damped oscillation with exponential decay time constant and oscillation frequency expressed as

$$\tau_{\rm dec} = \frac{2 \cdot L_{ds} \| L_{qs}}{R_s} = \frac{2 L_{ds} L_{qs}}{R_s (L_{ds} + L_{qs})}$$
$$f_{\rm osc} = \sqrt{\omega_r^2 - \frac{1}{\tau_{\rm dec}^2}} / 2\pi. \tag{8}$$

In the case of the experimental PMSM specified in Table II, $\tau_{\rm dec}$ is 26 ms. This time constant causes significant time delay in the settling of the current. If the voltage angle is rapidly changed, the current oscillation lasts for about five times of $\tau_{\rm dec}$, i.e., 130 ms. Since instantaneous torque is determined by instantaneous $d-q$ currents, the current oscillation leads to time delay of torque control. Even if the differential of θ is intentionally bounded to avoid the oscillation, that limitation itself would confine the dynamic performance of current control. The aforementioned limitations are coming from the deactivation of the current regulator. For better dynamic performance of the current control, the closed-loop current regulator should be maintained even during the six-step mode.

III. Voltage Synthesis by Overmodulation

Here, conventional overmodulation methods are minutely reviewed from the perspective of fundamental voltage synthesis to check their suitability to the six-step operation. Overmodulation technique is very important since it directly determines the quality of the inverter output during nonlinear modulation. The voltage constraint of the inverter output can be expressed as a voltage hexagon whose magnitude depends on DC-link voltage. Fig. 2(a) shows the voltage hexagon on a stationary voltage plane. With regard to the linearity of the voltage synthesis, the voltage plane can be divided into linear and nonlinear regions, as shown in Fig. 2(b). If the voltage reference vector is rotating in the linear region, the inverter can synthesize exactly the same voltage as the voltage reference in the moving average sense for a pulsewidth modulation (PWM) period. However, if the voltage reference goes beyond the voltage hexagon, dynamic overmodulation is carried out, and a point on the voltage hexagon is assigned for the inverter output voltage.

Fig. 3 shows three well-known overmodulation methods. Minimum distance overmodulation and switching-state overmodulation are equivalent to space vector PWM (SVPWM) and

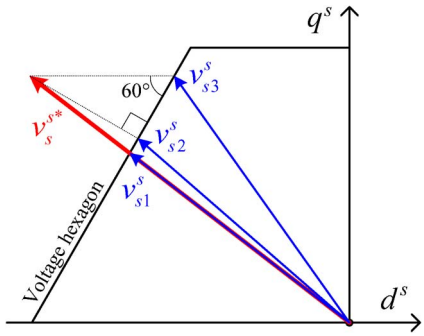


Fig. 3. Voltage assignments by conventional overmodulation methods. (v_s^{s*}: voltage reference; v_{s1}^s: minimum phase error overmodulation; v_{s2}^s: minimum distance overmodulation; v_{s3}^s: switching state overmodualtion).

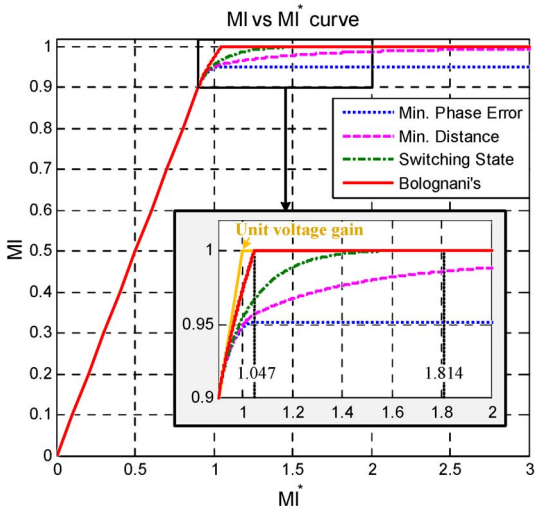


Fig. 4. MI versus MI* curve.

discontinuous PWM in nonlinear region, respectively. Modulation index (MI) can be defined as follows:

$$\text{MI} = \frac{V_{s1}}{\frac{2}{\pi} \cdot V_{\rm dc}}. \tag{9}$$

The denominator of (9), i.e., $2/\pi V_{\rm dc}$, is the maximum value of V_{s1} accomplished by the six-step operation. Fig. 4 shows the relation of MI and MI*, where MI* indicates the magnitude of circularly rotating voltage reference, i.e., V_s^*, divided by $2/\pi V_{\rm dc}$. Detailed calculation of MI is mostly covered in [7]. Until MI* reaches 0.907, SVPWM is used, and MI is identical to MI*. With MI* further increasing over 0.907, the increase of MI depends on the overmodulation method itself. For the implementation of the six-step operation, MI should reach to the unity at a certain value of MI*. Using minimum phase error overmodulation, MI is saturated at 0.952 when MI* reaches 1.047 ($V_s^* = 2/3 \cdot V_{\rm dc}$), which means that this method is unsuitable for the six-step operation. Since [18] adopted minimum phase error overmodulation, 4.8% of the inverter capability would be left unused. Using minimum distance overmodulation, MI gradually increases with MI* in nonlinear region. References [14]–[17] achieved high MI above 0.98 in the flux-weakening region by this overmodulation. However, MI does not actually reach to the unity even if MI* goes to

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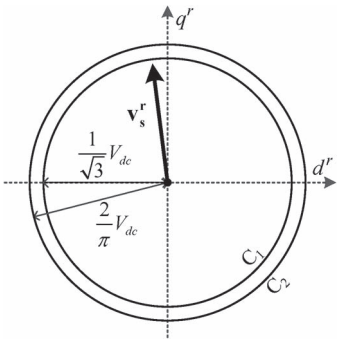


Fig. 5. Circle constraint of inverter output.

infinity, which means that minimum distance overmodulation is also unsuitable for the six-step operation. Switching-state overmodulation can be used for the six-step operation since MI reaches to the unity after MI* passes 1.814. However, such a nonlinearity of voltage gain, i.e., MI/MI*, varying from 1 to 0.551 could bring about significant degradation of dynamic performance during the transition from the linear modulation to the six-step mode. An overmodulation method with a larger voltage gain is preferred to keep the dynamic performance of the current regulator.

There have been efforts to keep the unit voltage gain in the overmodulation range [5]–[7]. Reference [5] proposed a method realizing the unity voltage gain based on Fourier series representation of the inverter output voltage. In [7], the authors proposed an inverse voltage gain method. With precalculated inverse voltage gain, the voltage reference was boosted by multiplying the inverse voltage gain before overmodulation process. However, their nature was a static overmodulation, which only focuses on per-fundamental-cycle voltage gain linearity. From the view point of dynamic overmodulation [19], they sometimes do undesired correction of the voltage reference. When the magnitude of the voltage reference is between $1/\sqrt{3}\cdot V_{\rm dc}$ and $2/3\cdot V_{\rm dc}$, they correct the voltage reference to maintain the unit voltage gain even if the voltage reference is inside the voltage hexagon where the correction is unnecessary. Considering dynamic operation, the voltage reference should be corrected only when it is out of the voltage hexagon.

IV. Tradeoff Relation Between Steady-State Voltage Utilization and Dynamic Performance

Basic current and voltage constraints can be expressed as

$$i_{ds}^{r2}+i_{qs}^{r2}\le I_{\rm lim}^2\cdot V_{s1}\le V_{\rm lim}. \tag{10}$$

$I_{\rm lim}$ is the current limit determined by the thermal capacity of PMSM drive system. In addition, $V_{\rm lim}$ indicates the limiting value of V_{s1}. The setting of $V_{\rm lim}$ is very tricky because it determines tradeoff relation between steady-state voltage utilization and dynamic performance of current control. In the sense of fundamental voltage synthesis, the constraint of inverter output can be described as a circle on a synchronous voltage plane. Fig. 5 shows the circle constraint of inverter output, where $\mathbf{v_s^r}$ indicates the inverter output voltage vector fed to the PMSM. In the figure, C_1 is the boundary of the linear region, and C_2 is the maximum boundary achieved by the six-step operation.

In applications where DC-link voltage utilization is not a concern, only the linear region was considered for the steady-state operation [1], [2]. In this case, $V_{\rm lim}$ is normally set as $0.52\cdot V_{\rm dc}$, with a 10% margin. Then, steady-state back EMF is controlled to be inside C_1. The area between C_1 and C_2 is utilized only in dynamic condition. For example, if step command for increasing torque is applied, voltage reference temporarily goes out of C_1. Then, the area between C_1 and C_2 is utilized by dynamic overmodulation. Because the nonlinear region is reserved as a voltage margin, dynamic performance is quite satisfactory.

If $V_{\rm lim}$ is set larger than $1/\sqrt{3}\cdot V_{\rm dc}$ for enhanced voltage utilization [14]–[18], back EMF is located between C_1 and C_2, and the nonlinear region is utilized even in the steady state. Although the steady-state voltage utilization increases, there is a relatively smaller voltage margin for the current control. As a result, dynamic performance of current control is significantly degraded.

The six-step operation is an extreme case that maximizes DC-link voltage utilization. $V_{\rm lim}$ is set to $2/\pi\cdot V_{\rm dc}$ and $\mathbf{v_s^r}$ is on C_2 during the steady-state operation. Because the inverter output is already maximized, there is no voltage margin reserved for transient-state current control. In other words, there is no way to increase the output voltage of the inverter even for a very short time. For this reason, maintaining current control performance is a challenging task.

V. Proposed Scheme

To implement the six-step operation of PMSM with closed-loop current control, a control scheme, which is described in Fig. 6, is proposed. The proposed scheme consists of three controllers: Part I, Part II, and Part III. In the proposed scheme, the six-step operation is implemented by Bolognani's overmodulation [4] in Part I, not by the voltage angle control. During the six-step operation, the voltage reference is kept rotating outside the voltage hexagon. Then, Part I would continuously produce six-step voltage. Part II is the flux-weakening controller that limits the magnitude of the voltage reference and determines the current references. Part III is for enhancing the dynamic performance of the current regulator. The three controllers are explained in detail here.

A. Part I: Dynamic Overmodulation for the Six-Step Operation

In Part I, Bolognani's overmodulation technique [4] is adopted. Rather than realizing the voltage gain linearity, the voltage reference is corrected only when it is out of the voltage hexagon, which is more suitable to dynamic operation. This correction only requires simple geometrical calculation. Fig. 7 shows the voltage correction on the $d-q$ plane and its corresponding phase voltage by Bolognani's method. When the voltage reference goes beyond the voltage hexagon, a point on the voltage hexagon with $\|\mathbf{v_s^r}\|=\|\mathbf{v_s^{r*}}\|$ is selected for the inverter output. Since points near the vertices on the voltage hexagon are frequently utilized, MI rapidly increases in the nonlinear region, as shown in Fig. 4. The reduction in voltage

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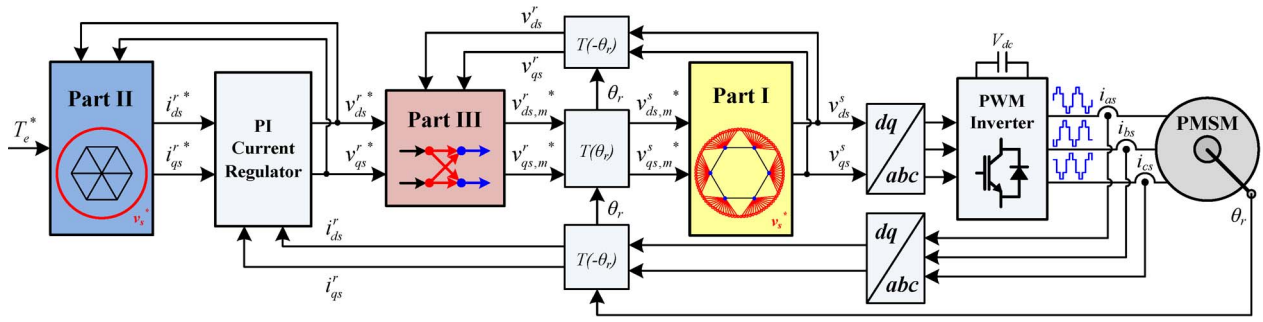


Fig. 6. Block diagram of the proposed drive system for six-step operation of PMSM.

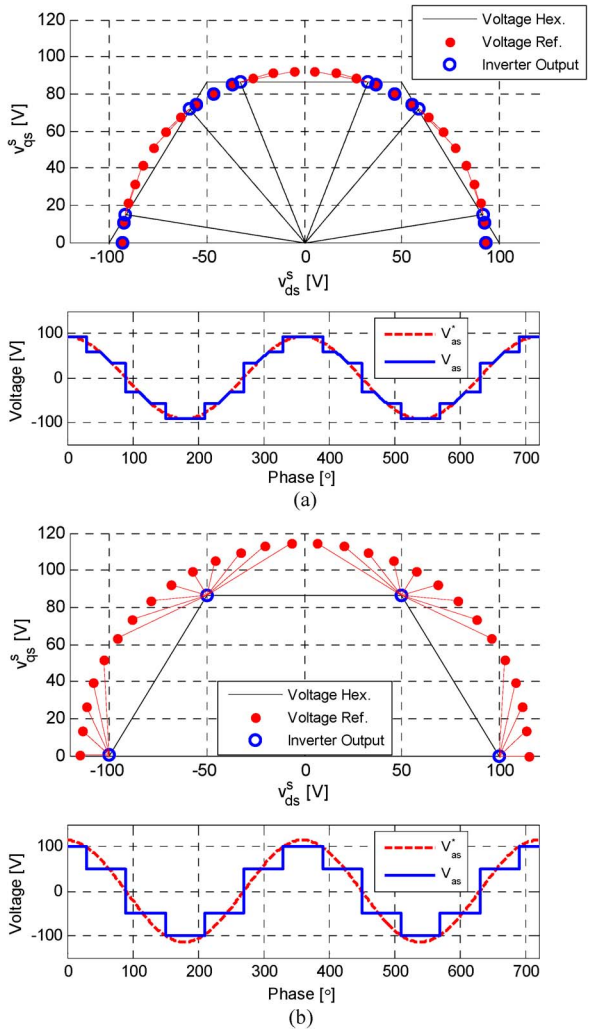


Fig. 7. Bolognani's overmodulation. (a) $\mathrm{MI}^* = 0.97(V_s^* = 0.618 \cdot V_{\mathrm{dc}})$. (b) $\mathrm{MI}^* = 1.2(V_s^* = 0.764 \cdot V_{\mathrm{dc}})$.

gain is less than 5%, and it can be compensated by closed-loop operation of controllers. If V_s^* is larger than $2/3 \cdot V_{\mathrm{dc}}$, the nearest vertex of the voltage hexagon is selected, realizing the six-step mode. In the proposed scheme, V_s^* is intentionally kept larger than $2/3 \cdot V_{\mathrm{dc}}$ by Part II to maintain the six-step mode.

B. Part II: Flux-Weakening Controller

The role of the flux-weakening controller is to determine the current references satisfying voltage and current constraints

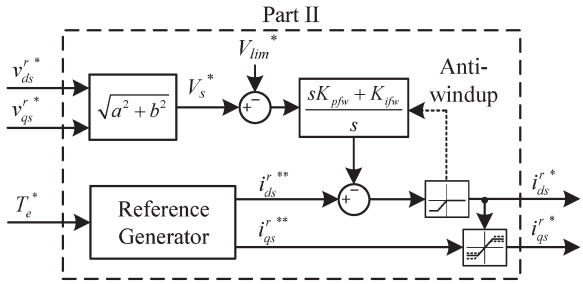


Fig. 8. Structure of Part II.

in (10). Because model-equation-based feedforwarding methods [1] are prone to parameter variation, only feedback-based methods [2], [14]–[18] are considered for the flux-weakening control. Fig. 8 shows the structure of Part II using PI regulator, where V_{lim}^* is the limiting value of V_s^*. PI-regulator-based flux-weakening controllers [2] have been widely used in industries for their structural simplicity and reasonable performance. However, in Part II, V_{lim}^* is set larger than $2/3 \cdot V_{\mathrm{dc}}$ for the six-step operation, whereas conventionally, it has been set smaller than $1/\sqrt{3} \cdot V_{\mathrm{dc}}$. When V_s^* is between $1/\sqrt{3} \cdot V_{\mathrm{dc}}$ and V_{lim}^*, overmodulation is performed by Part I without flux-weakening operation. When V_s^* hits V_{lim}^*, the PI regulator in Part II starts to operate, and V_s^* is regulated to V_{lim}^*. Antiwindup of the current regulator is intentionally deactivated to avoid steady-state error of the currents during flux-weakening operation [17].

Comparing Part II with LPF-based flux-weakening controllers [14]–[17], it has a merit that V_s^* can be regulated within a certain value, i.e., V_{lim}^*. Since MI depends on V_s^*, V_s^* should be kept within a certain value to get the desired MI. For instance, to achieve the six-step operation by Part I, further increasing V_s^* from $2/3 \cdot V_{\mathrm{dc}}$ does not have any effect on MI because it is already saturated at $V_s* = 2/3 \cdot V_{\mathrm{dc}}$. By the integrator in Part II, the error between V_s^* and V_{lim}^* can be eliminated during flux-weakening operation, whereas V_s^* varies depending on operating conditions by LPF-based flux-weakening controllers. In practical implementation, V_{lim}^* is set to $1.3 \cdot V_{\mathrm{dc}}$. Due to the ripples in V_s^* induced by current harmonics, V_{lim}^* is set larger than $2/3 \cdot V_{\mathrm{dc}}$, considering margin.

Fig. 9 shows capability curves from different flux-weakening methods. The curve in Fig. 9(a) can be achieved by normal flux-weakening control without overmodulation. In the figure, ω_{b1} and $\omega_{\max 1}$ indicate the base speed and the maximum speed, respectively. In Fig. 9(b), the capability curve by the proposed scheme is shown. The region denoted by OVM can be achieved

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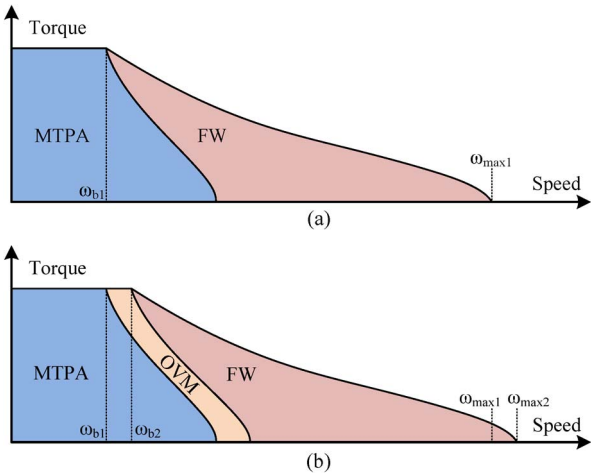


Fig. 9. Capability curves from different flux-weakening methods. (a) Flux-weakening control by the conventional scheme. (b) Flux-weakening control by the proposed scheme.

TABLE I
FEATURES OF REGIONS IN FIG. 9(b)

Name of region	MTPA	OVM	FW
MTPA control	O	O	X
Flux-weakening	X	X	O
Overmodulation	X	O	O
Six-step operation	X	X	O
MI	0~0.907	0.907~1	1

by overmodulation and MTPA control without flux-weakening. In the FW region, the PMSM is run by the six-step operation and the flux-weakening control. The features of each region in Fig. 9(b) are summarized in Table I.

C. Part III: Voltage Reference Modification for Improved Dynamic Performance

As explained in Section IV, instantaneous current control is very difficult during the six-step operation due to absolutely no voltage margin. The role of Part III is improving the dynamic performance of current control in this condition. Considering transient-state behavior of the currents, the resistive voltage drop term in (1) could be neglected. Then, the voltage equation can be approximated as

$$\begin{aligned} sL_{ds}i_{ds}^r &\approx v_{ds}^r - e_{ds}^r \\ sL_{qs}i_{qs}^r &\approx v_{qs}^r - e_{qs}^r \end{aligned} \tag{11}$$

$$\begin{aligned} e_{ds}^r &= -\omega_r L_{qs} i_{qs}^r \\ e_{qs}^r &= \omega_r \left(L_{ds} i_{ds}^r + \lambda_f\right). \end{aligned} \tag{12}$$

In (12), e_{ds}^r and e_{qs}^r are the $d-q$ components of effective back EMF coming from the rotation of the PMSM. Equation (11) implies that the variation of current is determined by the deviation between terminal voltage and effective back EMF. When the output torque of the motor should be rapidly increased, there should be a large difference between v_{qs}^r and e_{qs}^r to increase q-axis current. Since V_{s1} is fixed to $2/\pi V_{\rm dc}$ during the six-step operation, the only way to increase the gap between v_{qs}^r and e_{qs}^r is decreasing e_{qs}^r. As shown in (12), e_{qs}^r can be reduced by decreasing i_{ds}^r. In the same way, for increasing i_{ds}^r, e_{qs}^r should be decreased by increasing i_{qs}^r. Employing cross-coupled features between $d-q$ currents, dynamic performance can be remarkably improved.

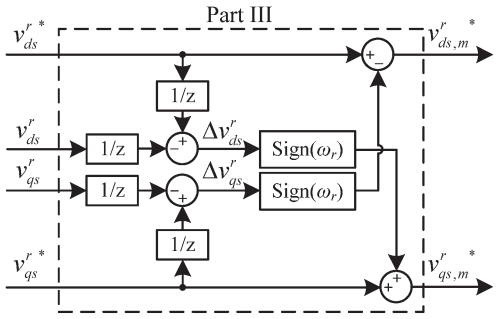


Fig. 10. Structure of Part III.

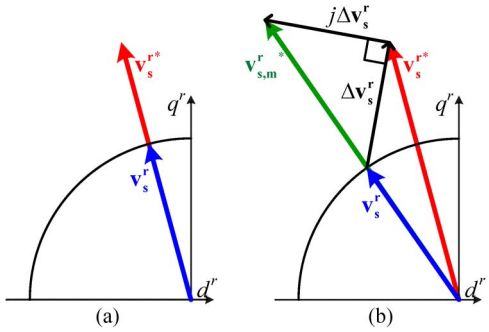


Fig. 11. Adjustment of voltage reference vector. (a) Without Part III. (b) With Part III.

Fig. 10 shows the implementation of Part III. In Part III, v_{ds}^{r*} and v_{qs}^{r*} from the current regulator are modified by

$$\begin{aligned} v_{ds,m}^{r*}[n+1] &= v_{ds}^{r*}[n+1] - \text{sign}(\omega_r)\cdot\Delta v_{qs}^r[n] \\ v_{qs,m}^{r*}[n+1] &= v_{qs}^{r*}[n+1] + \text{sign}(\omega_r)\cdot\Delta v_{ds}^r[n]. \end{aligned} \tag{13}$$

In (13), Δv_{ds}^r and Δv_{qs}^r indicate voltage errors between the voltage reference and the inverter output. Fig. 11 describes how the voltage vector is modified by Part III. The cross-coupling operation in (13) leads to

$$\mathbf{v_{s,m}^{r*}} = \mathbf{v_s^{r*}} + j\Delta\mathbf{v_s^r}. \tag{14}$$

That is, a 90° rotated voltage error vector is added to the original voltage reference vector. By the modification of the voltage reference, the actual inverter output voltage is modified, as shown in Fig. 11(b). This modification is closely related with the current regulator. Using the PI current regulator with back EMF feedforwarding, the output of the regulator is given by

$$\begin{aligned} v_{ds}^{r*} &= \frac{sK_{pd} + K_{id}}{s}\Delta i_{ds}^r + e_{ds}^r \\ v_{qs}^{r*} &= \frac{sK_{pq} + K_{iq}}{s}\Delta i_{qs}^r + e_{qs}^r. \end{aligned} \tag{15}$$

In (15), Δi_{ds}^r indicates the d-axis current error, i.e., $i_{ds}^{r*} - i_{ds}^r$. It can be inferred from (15) that the current regulator increases the

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TABLE II
EXPERIMENTAL SETTINGS AND PARAMETERS OF THE PMSM DRIVE SYSTEM

V_{dc}	150V	P	6 poles
Switching frequency	5kHz	R_s	0.15Ω
Sampling time	100μs	λ_f	0.254V·s
Current control BW	500Hz	L_{ds}	3.6mH
Rated current	39.5A_{rms}	L_{qs}	4.3mH

voltage references to reduce the current errors. In the voltage references, high-frequency components of the current errors are mostly dominated by the proportional parts. Thus, the following equations hold for high-frequency components of Δi_{ds}^r and y:

$$v_{ds}^{r*} \approx K_{pd}\Delta i_{ds}^r + e_{ds}^r$$
$$v_{qs}^{r*} \approx K_{pq}\Delta i_{qs}^r + e_{qs}^r. \quad (16)$$

Combining (16) with (11), voltage error terms can be expressed as

$$\Delta v_{ds}^r \approx K_{pd}\Delta i_{ds}^r - sL_{ds}i_{ds}^r$$
$$\Delta v_{qs}^r \approx K_{pq}\Delta i_{qs}^r - sL_{qs}i_{qs}^r. \quad (17)$$

In (17), the last term on the right side is the inductive voltage drop term, which induces the actual variation of the current. Then, the voltage error terms in (17) can be interpreted as an unmanaged portion of current error. In (13), the voltage error terms are used as indicators that show current errors unmanaged by the current regulator. By the cross-coupling operation of (13), the voltage references are dynamically adjusted from the current error state. Since Part III operates only when the voltage reference is outside the voltage hexagon, it does not influence steady-state operation under the base speed.

VI. SIMULATION RESULTS

In all simulations, system and control parameters in Table II are used. Three conditions are simulated by MATLAB Simulink.

A. Condition 1: Performance Evaluation of Part III

This condition is designed to evaluate the effectiveness of Part III independent of Part I and Part II. Part II is taken out, and Part I is replaced by the conventional minimum distance overmodulation method. Holding speed at 750 r/min, maximum torque command is applied. Fig. 12 shows simulation results. Fig. 12(a) and (b) shows current waveforms, and Fig. 12(c) and (d) shows voltage trajectories on the $d-q$ plane. In Fig. 12(a), the settling time of current is 11 ms. However, the settling time is reduced to 7 ms by Part III, as shown in Fig. 12(b). This 36% reduction in settling time comes from the voltage modification. Fig. 12(d) shows that when the voltage reference goes beyond the voltage hexagon, Part III rapidly modifies the voltage reference vector. It is worth noting that the d-axis current is rapidly decreased even under its reference by the voltage modification, and it helps in the regulation of the q-axis current. References [17] and [20] provided meaningful explanations of this phenomenon by qualitative and quantitative approaches, respectively.

B. Condition 2: Transition Between SVPWM and Six-Step Mode

The transition between SVPWM and six-step mode is simulated. Holding speed at 1000 r/min, maximum torque command is applied from 0.1 to 0.3 s. Fig. 13 shows the results. From 0 to 0.1 s, the operating point is located in the MTPA region in Fig. 9(b), and SVPWM is used for the current control. From 0.1 to 0.3 s, the operating point moves to the FW region in Fig. 9(b). In this period, the voltage synthesis method is changed to the six-step mode. Note that this seamless transition is not by algorithm switch but by continuous operation of the proposed controllers. The harmonic components of $d-q$ currents come from increased harmonics by the six-step operation. Although the preset bandwidth of the current regulator is 500 Hz, the current regulator is not severely excited by the harmonic current. This is because the output of the current regulator is saturated by the voltage hexagon. Its effective bandwidth during the six-step operation is much lower than 500 Hz. Anyway, the moving averages of $d-q$ currents in a PWM period are well regulated during the six-step operation. From 0.3 to 0.5 s, the torque reference is changed back to zero, and SVPWM is applied again.

C. Condition 3: Comparison of the Proposed Scheme and Conventional Schemes

In this condition, the step response of the proposed method is compared with that of the voltage angle control and the LPF-based flux-weakening control [14]–[17]. Holding speed at 1500 r/min, maximum torque command is applied from 0.1 to 0.3 s in a step manner. Fig. 14 shows the results. Fig. 14(a) shows the result from the voltage angle control. In this case, the six-step voltage is directly generated from the torque reference based on the steady-state torque equation (5). Although steady-state torque can be controlled, undesirable oscillation occurs in the currents. The amplitude of the oscillating component is exponentially decaying with time constant τ_{dec} in (8). Due to this oscillation, the settling time of current regulation is larger than 130 ms. Fig. 14(b) shows the result from the LPF-based flux-weakening controller in [14]. The currents are well controlled due to the simultaneous operation of the flux-weakening controller and the current regulator. As its voltage waveforms show, high MI is achieved by overmodulation. However, MI is less than the unity. One more thing to bring up from the voltage waveform is that MI is not kept within a certain value. As denoted on voltage waveform, MI in the maximum torque command period is higher than those in the other periods. This is because the radius of the voltage reference vector is not maintained using LPF. Fig. 14(c) shows the result from the proposed method. Among the three schemes, the proposed one shows the fastest regulation of the currents. The settling time of the q-axis current is about 10 ms when increasing torque. Although the d-axis current has a relatively longer settling time, it does not contribute much to the output torque. Its voltage

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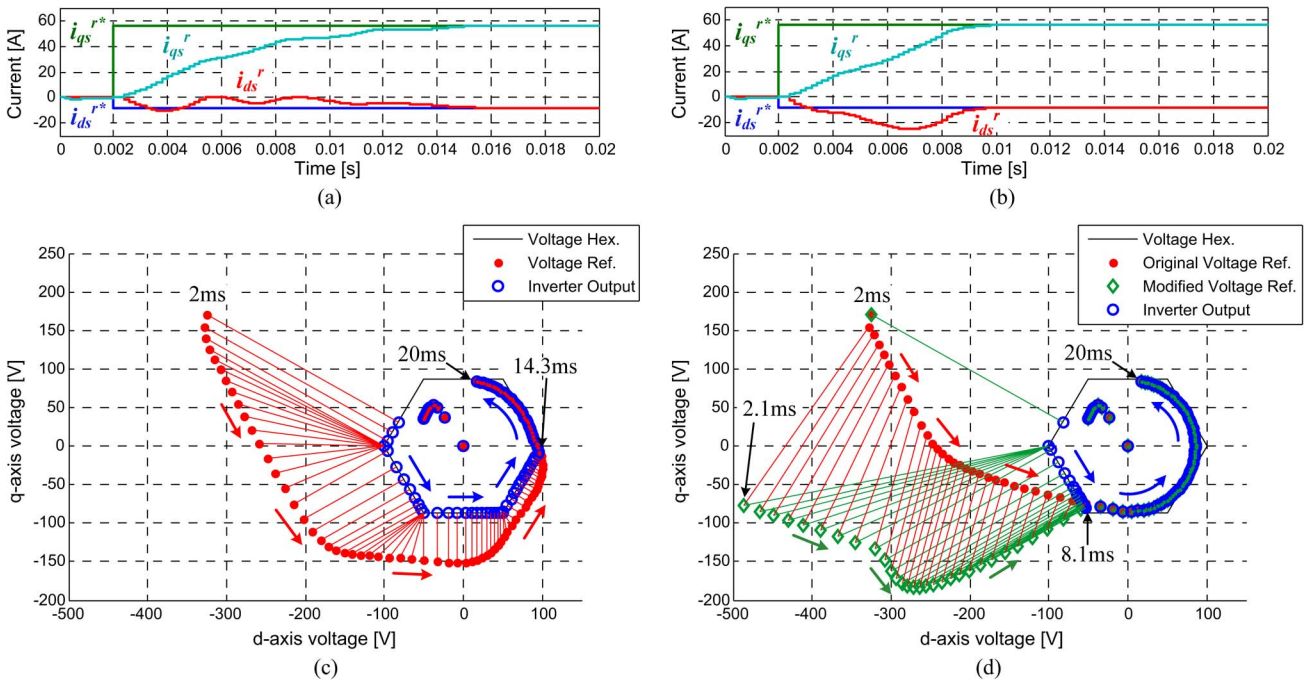


Fig. 12. Simulation results in condition 1. (a) Current waveform without Part III. (b) Current waveform with Part III. (c) Voltage trajectory without Part III. (d) Voltage trajectory with Part III.

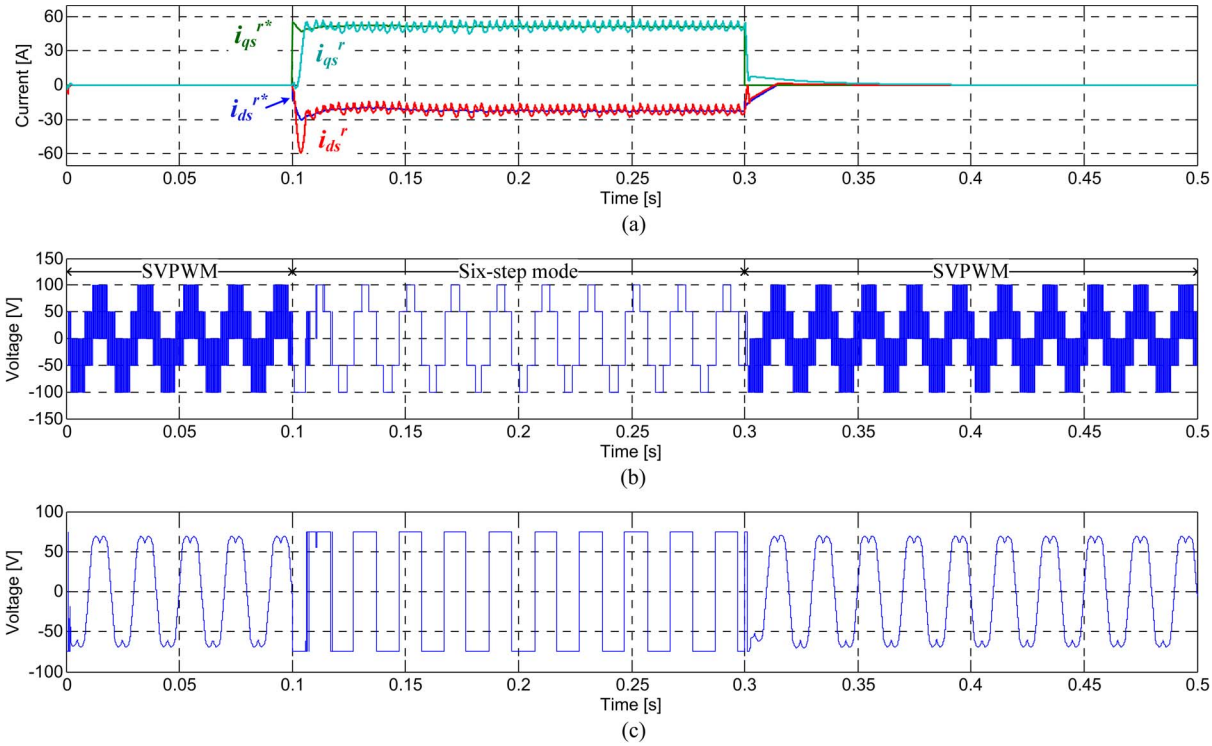


Fig. 13. Simulation results in condition 2. (a) $d-q$ currents. (b) A-phase voltage (v_{as}). (c) A-pole voltage reference (v_{an}).

waveform proves the realization of perfect six-step operation, except in transient period from 0.3 to 0.325 s, where the voltage reference vector goes inside the linear region for decreasing torque.

Fig. 15 shows voltage trajectories within a single electrical cycle in the maximum torque command period. The operation of minimum distance overmodulation is shown in Fig. 15(a). The ripples in the voltage reference come from the current harmonics induced by overmodulation. In Fig. 15(b), the voltage trajectory from the proposed scheme is shown. The voltage reference is successively adjusted by Part III, and overmodulation is done from the modified reference. Different from Fig. 15(a), only six vertices of the voltage hexagon are utilized for the inverter output, which leads to the six-step operation.

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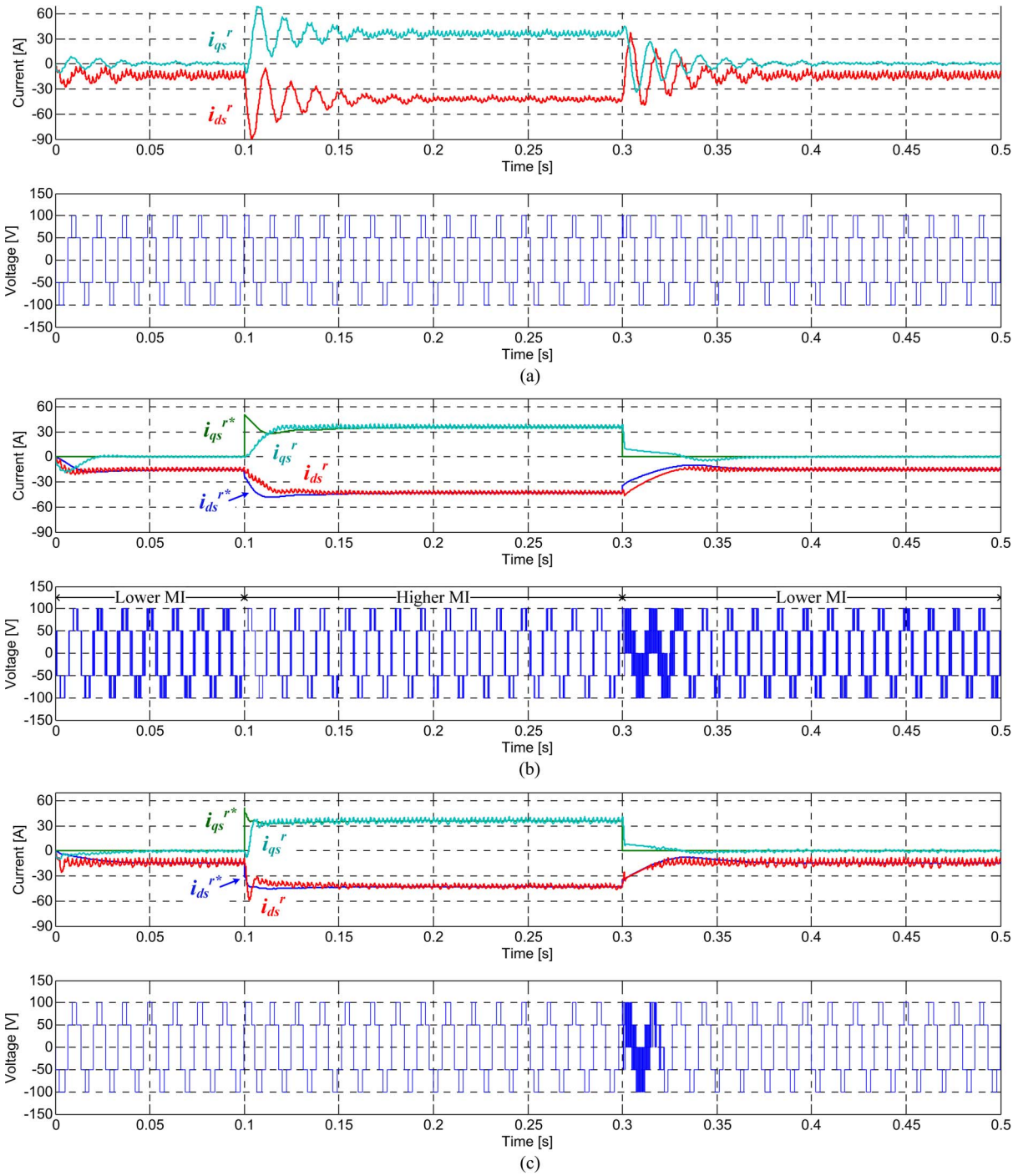


Fig. 14. Simulation results in condition 3. (a) $d-q$ currents and A-phase voltage from the voltage angle control. (b) $d-q$ currents and A-phase voltage from the LPF-based flux-weakening control. (c) $d-q$ currents and A-phase voltage from the proposed scheme.

VII. Experimental Results

Experimental settings and parameters are specified in Table II. Several tests are done to verify the effectiveness of the proposed scheme.

A. Test 1

The experimental test condition is the same as simulation condition 2. Speed is held at 1000 r/min by the loads machine, and maximum torque command is applied from 0.1 to 0.3 s. The voltage waveform in Fig. 16(b) is computed from measured line-to-line voltages, by $v_{as} = (v_{ab} - v_{ca})/3$. Experimental waveforms in Fig 16 well match to the simulation waveforms in Fig. 13. The increase in the amplitude of the phase voltage from 0.3 to 0.4 s comes from increased DC-link voltage by regenerative braking.

B. Test 2

The condition of this test is the same as simulation condition 3. Speed is held at 1500 r/min, and maximum torque

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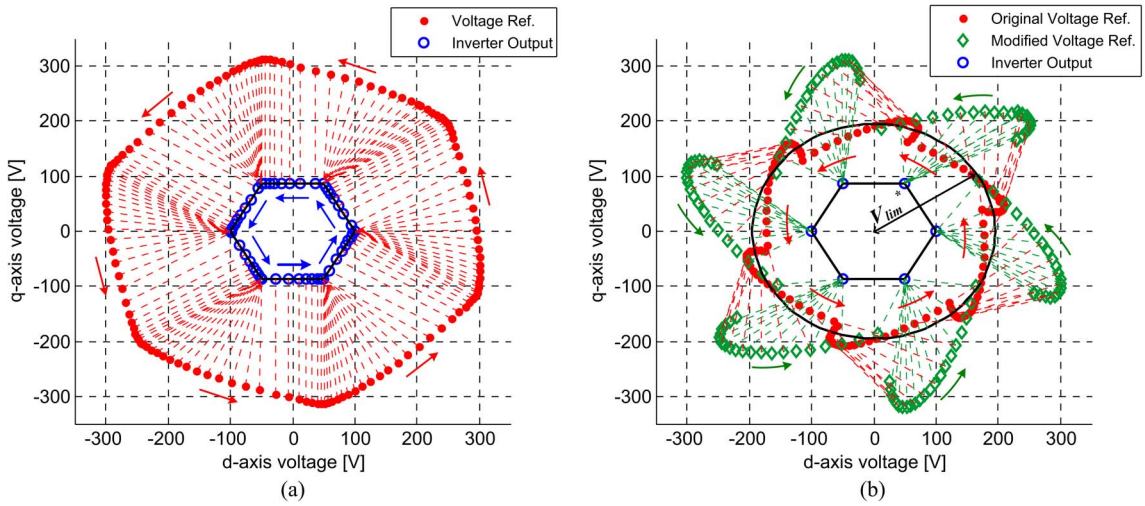


Fig. 15. Voltage trajectory in maximum torque command period in Fig. 14(a). (a) LPF-based flux-weakening control. (b) Proposed scheme.

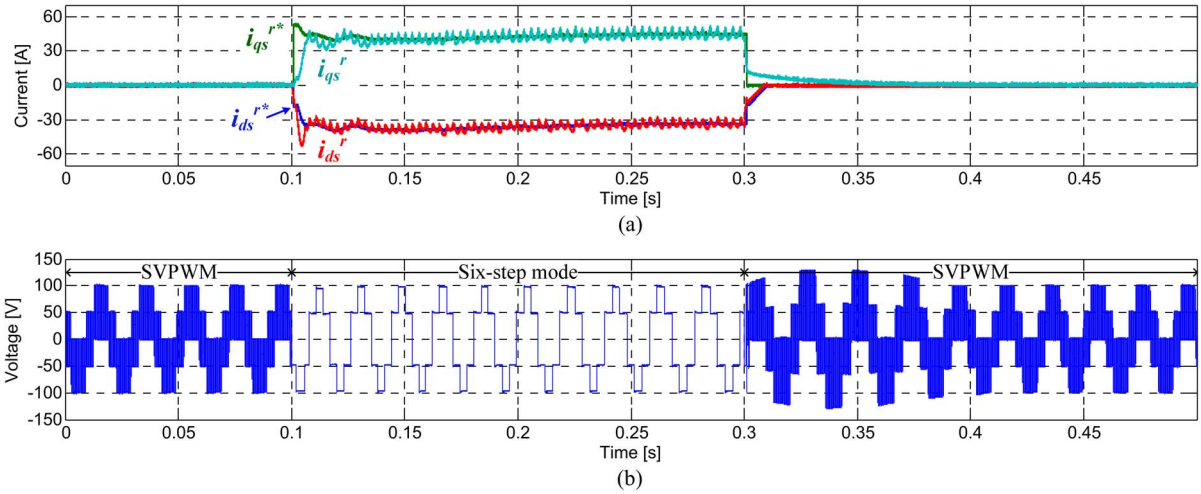


Fig. 16. Experimental test 1. (a) $d-q$ currents. (b) A-phase voltage (v_{as}).

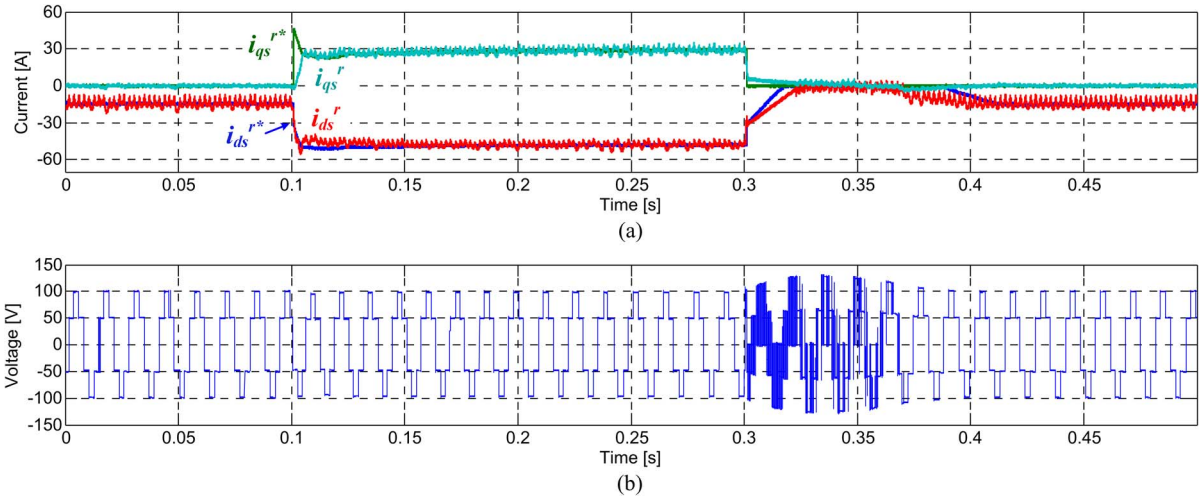


Fig. 17. Experimental test 2. (a) $d-q$ currents. (b) A-phase voltage (v_{as}).

command is applied from 0.1 to 0.3 s. Experimental waveforms in Fig. 17 match to the simulation waveforms in Fig. 14(c). Because flux-weakening control is needed at 1500 r/min even with zero torque command condition, the six-step mode is activated from 0 to 0.5 s, except in transient period from 0.3 to 0.37 s.

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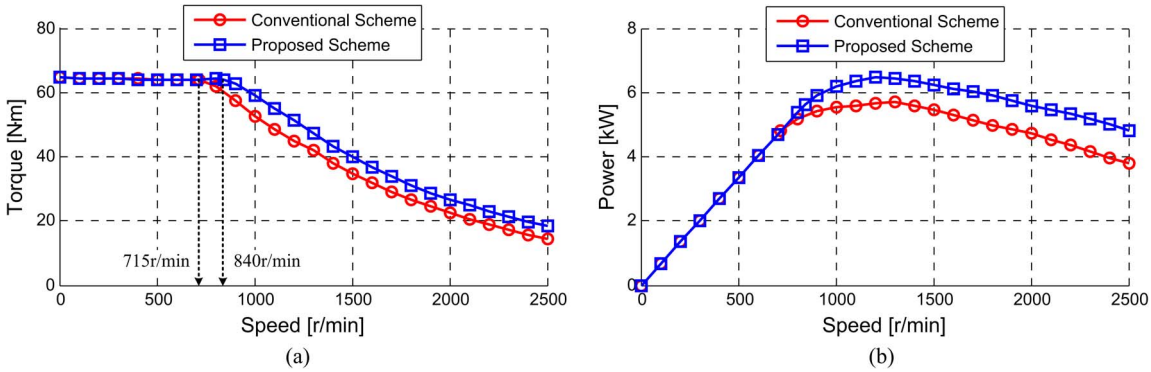


Fig. 18. Experimental test 3, capability curves. (a) Torque versus speed curve. (b) Power versus speed curve.

C. Test 3

In this test, steady-state torque is measured from zero speed to 2500 r/min experimentally. The conventional scheme indicates the scheme only utilizing the linear region [2] with a 5% voltage margin. The results are shown in Fig. 18. By the six-step operation, the maximum torque capability of the PMSM is achieved. The base speed is changed from 715 to 840 r/min, which means that the constant torque region has been extended by 17%. In addition, as shown in Fig. 18(b), the maximum power of the motor under the same current and voltage constraints has been enhanced by 14%.

VIII. Conclusion

This paper has covered investigations and realization of the six-step operation of PMSM with instantaneous current control. Applying the proposed method, the capability of PMSM can be enhanced up to the real physical limit. Furthermore, the instantaneous current control is achieved by continuous operation of the proposed controllers without mode switching. Computer simulation and experimental results support the validity of the proposed method. Even with stepwise change of the torque reference, $d-q$ currents are quickly regulated. At three times of the base speed, i.e., 2500 r/min, the torque capability is enhanced by 27% by the proposed method.

References

[1] S. Morimoto, Y. Takeda, and T. Hirasa, "Expansion of operating limits for permanent magnet motor by current vector control considering inverter capacity," *IEEE Trans. Ind. Appl.*, vol. 26, no. 5, pp. 866–871, Sep./Oct. 1990.

[2] J. M. Kim and S. K. Sul, "Speed control of interior permanent magnet synchronous motor drive for the flux weakening operation," *IEEE Trans. Ind. Appl.*, vol. 33, no. 1, pp. 43–48, Jan./Feb. 1997.

[3] J. Holtz, W. Lotzkat, and M. Khambadkone, "On continuous control of PWM inverters in the overmodulation range including the six-step mode," *IEEE Trans. Power Electron.*, vol. 8, no. 4, pp. 546–553, Oct. 1993.

[4] S. Bolognani and M. Zigliotto, "Novel digital continuous control of SVM inverters in the overmodulation range," *IEEE Trans. Ind. Appl.*, vol. 33, no. 2, pp. 525–530, Mar./Apr. 1997.

[5] D. C. Lee and G. M. Lee, "A novel overmodulation technique for space-vector PWM inverters," *IEEE Trans. Power Electron.*, vol. 13, no. 6, pp. 1144–1151, Nov. 1998.

[6] T. H. Nguyen, T. L. Van, D. C. Lee, J. H. Park, and J. H. Hwang, "Control mode switching of induction machine drives between vector control and V/f control in overmodulation range," *J. Power Electron.*, vol. 11, no. 6, pp. 846–855, Nov. 2011.

[7] A. M. Hava, R. J. Kerkman, and T. A. Lipo, "Carrier-based PWM-VSI overmodulation strategies: Analysis, comparison, and design," *IEEE Trans. Power Electron.*, vol. 13, no. 4, pp. 674–689, Jul. 1998.

[8] R. Monajemy and R. Krishnan, "Performance comparison for six-step voltage and constant back EMF control strategies for PMSM," in *Conf. Rec. IEEE IAS Annu. Meeting*, 1999, vol. 1, pp. 165–172.

[9] S. Morimoto, Y. Inoue, T. F. Weng, and M. Sananda, "Position sensorless PMSM drive system including square-wave operation at high-speed," *in Conf. Rec. IEEE IAS Annu. Meeting*, 2007, pp. 676–682.

[10] B. K. Bose, "A high-performance inverter-fed drive system of an interior permanent magnet synchronous machine," *IEEE Trans. Ind. Appl.*, vol. 24, no. 6, pp. 987–1087, Nov./Dec. 1988.

[11] H. Nakai, H. Ohtani, E. Satoh, and Y. Inaguma, "Development and testing of the torque control for the permanent-magnet synchronous motor," *IEEE Trans. Ind. Electron.*, vol. 52, no. 3, pp. 800–806, Jun. 2005.

[12] K. Asano *et al.*, "High performance motor drive technologies for hybrid vehicles," in *Proc. PCC*, Nagoya, Japan, 2007, pp. 1584–1589.

[13] T. Schoenen, A. Krings, D. van Treek, and R. W. De Doncker, "Maximum DC-link voltage utilization for optimal operation of IPMSM," in *Proc. IEEE IEMDC*, 2009, pp. 1547–1550.

[14] T. S. Kwon and S. K. Sul, "Novel antiwindup of a current regulator of a surface-mounted permanent-magnet motor for flux-weakening control," *IEEE Trans. Ind. Appl.*, vol. 42, no. 5, pp. 1293–1300, Sep./Oct. 2006.

[15] T. S. Kwon, K. Y. Choi, M. S. Kwak, and S. K. Sul, "Novel flux-weakening control of an IPMSM for quasi-six-step operation," *IEEE Trans. Ind. Appl.*, vol. 44, no. 6, pp. 1722–1731, Nov./Dec. 2008.

[16] H. Liu, Z. Q. Zhu, E. Mohamaed, Y. Fu, and X. Qi, "Flux-weakening control of nonsalient pole PMSM having large winding inductance, accounting for resistive voltage drop and inverter nonlinearities," *IEEE Trans. Power Electron.*, vol. 27, no. 2, pp. 942–952, Feb. 2012.

[17] Y. C. Kwon, S. Kim, and S. K. Sul, "Voltage feedback current control scheme for improved transient performance of permanent magnet synchronous machine drives," *IEEE Trans. Ind. Electron.*, vol. 59, no. 9, pp. 3373–3382, Sep. 2012.

[18] P. Y. Lin and Y. S. Lai, "Voltage control technique for the extension of DC-link voltage utilization of finite-speed SPMSM drives," *IEEE Trans. Ind. Electron.*, vol. 59, no. 9, pp. 3392–3402, Sep. 2012.

[19] A. M. Hava, S. K. Sul, R. J. Kerkman, and T. A. Lipo, "Dynamic overmodulation characteristics of triangle intersection PWM methods," *IEEE Trans. Ind. Appl.*, vol. 35, no. 4, pp. 896–907, Jul./Aug. 1999.

[20] J. W. Choi and S. K. Sul, "Generalized solution of minimum time control in three-phase balanced systems," *IEEE Trans. Ind. Electron.*, vol. 45, no. 5, pp. 738–744, Oct. 1998.



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